

Element K: Reflection on Design Project

Defining The Problem (elements A-C)

Successes ● We clearly defined the core problem: creating a sustainable and effective bookend that supports the Green School Initiative. ● We gathered feedback from multiple stakeholders to ensure our goals were user focused.	Challenges ● The problem was quite broad so it took time to refine the design criteria and find out how to evaluate its efficiency.
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Ideation (element D)

Successes ● We explored multiple materials and conducted detailed research using reputable sources. ● We compared real world strengths and weaknesses to justify our eventual choice of wood as the ideal material.	Challenges ● While we brainstormed thoroughly, we initially learned not to consider the actual process of cutting and attaching the wood.
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Selection & Prototype Design (elements D-F)

Successes ● We selected wood and considered several test designs created by several of our team members. ● We were able to come to a consensus and compromise by combining several of the designs after a few votes to decide on the shape and size.	Challenges ● We again did not consider the order in which we should build, glue and create the actual prototype which led to some challenges.
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Prototype Building (element G)

Successes ● We successfully built a wooden prototype using the basic tools. ● We were able to use sandpaper to	Challenges ● We had to change the original design to cut the wooden block in a different shape because the original design was
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heavily sand down the wooden block to make sure it was even and smooth.	not the safest or easiest to cut. Some parts of the cut were imprecise due to limited tool availability which required us to spend a lot of time standing.
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Testing (elements H-I)

Successes - We tested the prototype in mimicked library conditions and gathered direct feedback from stakeholders. - Our bookend was successful in holding several books without falling over and was praised for being aesthetically pleasing and sustainable.	Challenges - In our test, the bookend was able to grip as well as expected but weight balance was not ideal for larger textbooks. - The base lacked non-slip materials which we couldn't add due to it becoming unbalanced. - The bottom cardboard covered with paint was quite useful for non-slip uses however.
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Evaluation (element J)

Successes - We used stakeholders feedback to effectively identify improvements - All evolutions confirmed that our design met most requirements: recyclable, aesthetic and sturdy	Challenges - We only evaluated one prototype in depth and did not get a chance to iterate further or explore variations - This limited some opportunity for improvement
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Lessons Learned About the Engineering Design Process

- Problem should be defined clearly and early
 - One major takeaway is that design constraints should be specific, stakeholder driven and realistic.
- Ideas should balance creativity and feasibility
 - Brainstorming should also involve being grounded in construction limitations and realistic expectations.
 - In future projects, we should prioritize ideas not only for their innovation but how implementable they are considering the resources and people involved
- Testing should be realistic and varied

- a. Bookends should be tested on different shelves especially considering their varied uses
- 4. Stakeholder feedback is necessary for strong evaluation
 - a. Without feedback, the improvements are not accurately representing the needs of the people using the constructed prototype
- 5. Documentation and Research are just as important as the build
 - a. The time spent researching and preparing is needed to ensure smarter decisions which will limit setbacks in the future

Individual Reflections

Aashi: The project was overall quite successful because we were able to create a good final prototype. We compromised when necessary and made sure everyone's opinions were heard to ensure a smooth execution of the prototype construction.

Rin: Our team had good collaboration and communication, and we had minimal disagreements. When we did disagree, we figured out a solution and assessed our options logically. We split our work fairly as well.

Stephen: I enjoyed the process of constructing the prototype the most. Step

Ahmed: Our team had good synergy with each other as we split the work up for more efficiency. Everyone made sure to keep everyone in the loop on this project as we were all caught up with each other's progress and every idea was heard and understood. Overall, this project was successful with this being a smooth, collaborative project.

Miracle: I enjoyed working with my team for this project and we were very teamwork-driven. We discussed each part from sketching to edits to building and we reviewed things with each other as we completed our elements. For most elements, we worked in pairs to complete them and this helped to ensure that everyone stayed up-to-date and connected to each other.

Faith: Our team worked together extremely well. It was a good idea to split the team into builders and portfolio workers. We assigned people to each element, and we were able to all get everything done on time. Our design was incredible and I'm very proud of our work, how well it works, how pretty it is, and the compliments from various stakeholders and experts. All in all, the project was a great success.

Zion: Overall I feel as though our project was very successful despite the setbacks we had during the process. Our team worked well together, everyone contributed in the building, design and elements, and together we were able to adapt our design and process to the challenges we came across. I am very proud of the way we were able to work together and create a final product that worked so well.